

HURON COUNTY AP, MI, USA

WMO: 725406

Lat: 43.780N

Lon: 82.986W

Elev: 234

StdP: 98.55

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 54817

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.1	-16.3	-22.5	0.5	-18.9	-19.5	0.7	-15.3	11.4	-4.0	10.6	-4.7	2.5	260	0.557

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.0	30.9	22.3	28.9	21.5	27.5	20.9	24.0	28.4	23.1	27.3	22.2	26.0	4.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.7	17.9	26.3	22.1	17.2	25.7	21.1	16.2	24.7	73.4	28.6	69.6	27.1	66.1	25.7	27.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.4	8.9	8.0			-23.8	33.1	4.5	1.4	-27.1	34.1	-29.7	34.8	-32.3	35.6
			DB		-24.0	25.9	4.5	1.0	-27.3	26.6	-29.9	27.2	-32.4	27.8
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.7	-5.7	-5.3	-0.2	6.4	12.7	17.8	20.0	19.2	15.8	9.5	3.5	-2.2
	DBStd	10.39	6.10	5.97	6.13	5.10	4.98	4.05	3.36	3.16	4.31	4.83	4.86	5.09
	HDD10.0	2051	487	429	321	134	26	0	0	0	4	69	202	379
	HDD18.3	4094	745	663	574	359	187	59	21	28	100	276	445	637
	CDD10.0	1213	0	0	6	26	111	234	312	286	178	55	7	1
	CDD18.3	214	0	0	1	1	13	43	74	55	24	3	0	0
	CDH23.3	2004	0	0	3	18	143	424	716	459	216	25	0	0
	CDH26.7	507	0	0	0	2	31	122	193	106	51	4	0	0
Wind	WSAvg	3.4	4.2	4.0	3.8	4.0	3.3	2.8	2.4	2.3	2.6	3.5	3.9	4.1
Precipitation	PrecAvg	814	52	50	46	74	86	80	79	76	81	71	61	56
	PrecMax	1030	97	151	114	148	178	176	141	127	187	142	141	109
	PrecMin	625	14	16	19	19	41	24	29	17	10	37	18	19
	PrecStd	101	23	30	25	32	38	38	28	29	39	26	32	24
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.1	12.5	21.2	26.1	29.8	32.2	32.7	32.3	30.9	27.0	18.9	13.8
		MCWB	10.4	9.7	15.9	16.6	19.4	22.4	23.9	23.0	21.9	20.1	13.9	12.4
	2%	DB	7.2	7.6	16.0	22.3	27.2	29.9	30.7	29.1	27.9	22.8	15.9	9.0
		MCWB	5.8	6.1	11.9	14.9	19.1	21.3	22.6	21.9	20.5	16.8	12.3	7.8
	5%	DB	4.0	4.5	12.2	18.8	24.4	27.7	28.8	27.5	26.1	20.1	13.0	6.4
		MCWB	2.7	2.9	9.3	12.9	17.7	20.8	21.8	21.5	19.6	16.2	10.1	5.1
	10%	DB	2.4	2.4	8.2	15.9	22.2	26.0	27.3	26.1	23.7	17.6	11.1	4.1
		MCWB	1.6	1.1	5.8	10.8	16.2	19.4	21.1	20.2	18.7	14.3	9.0	2.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.4	10.8	15.9	18.8	22.6	24.2	25.8	25.1	23.9	21.5	15.2	12.4
		MCDB	10.9	11.7	20.9	23.6	27.1	29.2	29.8	28.6	28.3	25.7	17.5	13.7
	2%	WB	6.1	6.4	12.4	15.8	20.6	23.1	24.4	23.7	22.4	18.7	12.8	8.0
		MCDB	6.9	8.3	15.1	20.3	25.0	28.1	28.7	27.7	26.0	21.4	15.2	8.7
	5%	WB	3.1	3.0	9.4	13.6	19.0	21.9	23.3	22.5	21.1	16.7	10.9	5.5
		MCDB	3.9	4.2	11.9	18.0	22.8	26.3	27.1	26.1	24.0	19.3	12.6	6.5
	10%	WB	1.5	1.1	6.1	11.2	17.2	20.6	22.2	21.4	19.7	14.8	8.8	3.1
		MCDB	2.2	2.2	8.3	15.0	21.1	24.2	25.9	24.4	22.7	17.4	10.9	4.1
Mean Daily Temperature Range		MDBR	7.5	8.3	8.9	10.7	11.7	11.9	12.0	11.6	12.0	9.8	7.8	6.1
	5% DB	MCDBR	8.2	9.6	13.6	16.0	14.3	13.7	13.6	12.7	13.9	13.3	11.2	8.3
		MCWBR	7.4	8.3	9.9	9.9	8.2	7.4	6.8	6.9	7.4	8.4	8.6	7.4
	5% WB	MCDBR	7.9	8.7	12.9	14.3	12.5	12.1	11.2	10.9	11.2	11.4	10.1	8.3
		MCWBR	7.3	7.8	10.1	10.0	8.2	7.5	6.6	6.5	6.7	8.1	8.6	7.7
Clear-Sky Solar Irradiance	taub	0.298	0.305	0.325	0.358	0.383	0.410	0.405	0.397	0.368	0.349	0.312	0.282	
	taud	2.389	2.352	2.384	2.341	2.337	2.285	2.329	2.337	2.414	2.442	2.508	2.488	
	Ebn at Noon	828	887	914	906	889	862	861	856	858	827	813	822	
	Edh at Noon	77	94	105	119	123	130	123	118	101	87	69	64	
All-Sky Solar Radiation	RadAvg	1.28	2.11	3.48	4.55	5.38	6.03	6.02	5.17	4.09	2.53	1.49	1.00	
	RadStd	0.14	0.25	0.36	0.39	0.46	0.34	0.28	0.23	0.39	0.25	0.20	0.10	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	-0.81	N/A	N/A	N/A	N/A
(47)	Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page